SEMTM0051 — Foundations of Artificial Intelligence

Neural networks

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9 Exercises

The exercise numbering is the same as in the lecture notes. References to earlier sections and equations refer to those notes.

The exercises begin with the dimensions of a network and the effect of composing its layers. We then practise a complete forward calculation and distinguish fixed features from features whose parameters are learned. We check why ReLU is not differentiable at zero and examine how initialisation affects gradient descent. The final two exercises construct a function using ReLU and distinguish correct class predictions from the probabilities assigned to those classes.

1. **Dimensions and parameters.** A network takes an input row $\boldsymbol{x} \in \mathbb{R}^{1\times 3}$, has one ReLU hidden layer of width 4, and outputs two logits. Both affine layers have biases.

   (a) State the shapes of both weight matrices, both biases, and the hidden preactivation and activation rows. Write $[\boldsymbol{a}_1(\boldsymbol{x})]_j$ in terms of the input coordinates, first-layer weights and bias.

   (b) Count the scalar parameters. How many would a single affine map from the same inputs to the same outputs have?

2. **Two affine layers.** Let

$$\boldsymbol{u}(\boldsymbol{x}) = \boldsymbol{x}\begin{pmatrix} 1 & 0 \\ 2 & -1 \end{pmatrix} + \begin{pmatrix} 1 & -1 \end{pmatrix}, \qquad v(\boldsymbol{z}) = \boldsymbol{z}\begin{pmatrix} 2 \\ 1 \end{pmatrix} + 3. \tag{48}$$

   (a) Compute the effective weight column and bias of $v \circ \boldsymbol{u}$. Evaluate the composition at $\boldsymbol{x} = (0, 1)$, both by the two stages and by the collapsed formula.

   (b) Now insert ReLU between the stages and define $F(\boldsymbol{x}) = v(\mathbf{ReLU}(\boldsymbol{u}(\boldsymbol{x})))$. Evaluate F at the same input.

   (c) To check whether F is affine, consider inputs $\boldsymbol{x} = (t, 0)$. Write $F((t, 0))$ as a piecewise-defined function of t. Use its slopes to determine whether F is affine.

3. **A new input to the worked network.** Use the parameters (17) with $\boldsymbol{x} = (2, 0)$.

   (a) Compute the hidden preactivations, hidden activations and scalar output.

   (b) Interpret the output as a regression prediction with target $y = 1$, and compute the squared loss.

   (c) Instead interpret it as a binary logit with target $y = 1$. Give the class-1 probability, predicted label and logistic loss.

4. **Fixed features and learned features.** Return to the same worked network, whose first-layer parameters are

$$\boldsymbol{W}_1 = \begin{pmatrix} 1 & 2 & -1 \\ -1 & 0 & 1 \end{pmatrix}, \qquad \boldsymbol{b}_1 = \begin{pmatrix} 0 & 1 & 0 \end{pmatrix}.$$

Consider the two inputs $\boldsymbol{x}_A = (-1, -1)$ and $\boldsymbol{x}_B = (-2, -2)$. We want regression outputs 1 and 0, respectively.

(a) Write the hidden feature row $\boldsymbol{\phi}_1(\boldsymbol{x})$ as a function of a general input $\boldsymbol{x} = (x_1, x_2)$, and evaluate it at $\boldsymbol{x}_A$ and $\boldsymbol{x}_B$.

(b) Keep $\boldsymbol{W}_1$ and $\boldsymbol{b}_1$ fixed. Can any choice of final-layer weights $\boldsymbol{W}_2$ and bias b_2 produce the two required outputs? Justify your answer for all choices of these final-layer parameters.

(c) Change the second entry of $\boldsymbol{b}_1$ from 1 to 3, keeping $\boldsymbol{W}_1$ and the other hidden biases unchanged. Denote the new hidden feature map by $\widetilde{\boldsymbol{\phi}}_1$. Evaluate it at both inputs and find one choice of $\boldsymbol{W}_2$ and b_2 that now produces the required outputs. Verify both predictions.

5. **ReLU at zero.** Compute the left and right derivatives of ReLU at zero using difference quotients, and determine whether an ordinary derivative exists there.

6. **Gradient descent from two initialisations.** Return to the regression problem for the network (42). Use the squared-loss formula in (44).

(a) Calculate $\mathcal{L}'_{\text{sq}}(\theta)$ at $\theta = -4, 0, 4$. For each point, use the sign of the derivative to state whether a gradient-descent update with $\eta > 0$ increases or decreases θ.

(b) For the same regression problem, perform two gradient-descent updates with $\eta = 0.2$, first starting from $\theta^{(0)} = -2$ and then starting from $\theta^{(0)} = 2$. In each run, compute the squared loss at $t = 0, 1, 2$. Use the update formula in each region to justify which minimum each run approaches. Give the limiting loss of each run, and use these two runs to test the claim that decreasing the loss at every update guarantees convergence to a global minimum.

7. **Constructing a function with ReLU.**

(a) Write

$$U(x) = 2\,\text{ReLU}(x + 1) - 4\,\text{ReLU}(x) + 2\,\text{ReLU}(x - 1) \tag{49}$$

as a piecewise-defined function, and find its value at $x = 0$.

(b) Give weights and biases of a one-hidden-layer network that represents U exactly.

8. **Correct labels, probabilities and loss.** Return to the four XOR observations and the network in (38).

(a) Keep the first layer fixed and give final-layer weights and bias that multiply the original score by a factor $\alpha > 0$. Denote the resulting parameter vector by $\boldsymbol{\theta}(\alpha)$. Compute the four scores, the predicted labels and the probability assigned to each observed label.

(b) Calculate the logistic loss of each observation and hence the total loss $\mathcal{L}_{\log}(\boldsymbol{\theta}(\alpha))$. Determine whether the total loss can be zero for any finite $\alpha > 0$, and calculate its limit as α tends to infinity.

Solutions

1. Dimensions and parameters.

(a) Shapes and entries. The first layer maps three input coordinates to four hidden preactivations. The second maps four hidden activations to two output scores. Therefore the shapes are

$$\boldsymbol{W}_1 \in \mathbb{R}^{3\times 4}, \quad \boldsymbol{b}_1, \boldsymbol{a}_1(\boldsymbol{x}), \boldsymbol{\phi}_1(\boldsymbol{x}) \in \mathbb{R}^{1\times 4},$$
$$\boldsymbol{W}_2 \in \mathbb{R}^{4\times 2}, \qquad \boldsymbol{b}_2 \in \mathbb{R}^{1\times 2}.$$

For hidden unit j, the first-layer calculation is

$$[\boldsymbol{a}_1(\boldsymbol{x})]_j = \sum_{k=1}^{3} x_k (\boldsymbol{W}_1)_{kj} + (\boldsymbol{b}_1)_j.$$

Thus $(\boldsymbol{W}_1)_{kj}$ multiplies input coordinate x_k and contributes to hidden preactivation $[\boldsymbol{a}_1(\boldsymbol{x})]_j$. We apply ReLU after adding these weighted inputs and the bias.

(b) Parameter counts. The first layer has $3 \cdot 4 = 12$ weights and four biases, giving 16 parameters. The second has $4 \cdot 2 = 8$ weights and two biases, giving another 10. The network therefore has 26 parameters in total. A single affine map from three inputs to two outputs would instead have $3 \cdot 2 + 2 = 8$ parameters.

2. Two affine layers.

(a) Collapsing the affine layers. Write the first matrix and bias row as $\boldsymbol{A}$ and $\boldsymbol{b}$, and the second weight column as $\boldsymbol{c}$. Substituting the first map into the second gives

$$v(\boldsymbol{u}(\boldsymbol{x})) = (\boldsymbol{x}\boldsymbol{A} + \boldsymbol{b})\boldsymbol{c} + 3 = \boldsymbol{x}(\boldsymbol{A}\boldsymbol{c}) + (\boldsymbol{b}\boldsymbol{c} + 3).$$

The effective weight column and bias are therefore

$$\boldsymbol{A}\boldsymbol{c} = \begin{pmatrix} 1 & 0 \\ 2 & -1 \end{pmatrix} \begin{pmatrix} 2 \\ 1 \end{pmatrix} = \begin{pmatrix} 2 \\ 3 \end{pmatrix}, \qquad \boldsymbol{b}\boldsymbol{c} + 3 = 2 - 1 + 3 = 4.$$

Consequently,

$$(v \circ \boldsymbol{u})(\boldsymbol{x}) = 2x_1 + 3x_2 + 4. \tag{50}$$

At $\boldsymbol{x} = (0, 1)$, the first stage gives

$$\boldsymbol{u}((0, 1)) = (3, -2),$$

and the second gives $2 \cdot 3 - 2 + 3 = 7$. The collapsed formula also gives $0 + 3 + 4 = 7$.

(b) Inserting ReLU. With ReLU inserted, the hidden row at $\boldsymbol{x} = (0, 1)$ becomes $(3, 0)$, so $F((0, 1)) = 2 \cdot 3 + 0 + 3 = 9$. Thus F differs from the affine function in part (a). This comparison alone does not establish that F is non-affine.

(c) Checking the change of slope. For $\boldsymbol{x} = (t, 0)$, the first affine layer gives $\boldsymbol{u}((t, 0)) = (t + 1, -1)$. Hence

$$F((t, 0)) = 2\,\mathrm{ReLU}(t + 1) + 3 = \begin{cases} 3, & t \le -1, \\ 2t + 5, & t > -1. \end{cases}$$

The slopes are zero for $t < -1$ and two for $t > -1$. If F were affine in (x_1, x_2), setting $x_2 = 0$ would give an affine function of t, with a constant slope. The change of slope therefore proves that F is not affine.

3. A new input to the worked network.

(a) Forward pass. The worked network has output-layer parameters

$$\boldsymbol{W}_2 = \begin{pmatrix} 2 \\ -1 \\ \frac{1}{2} \end{pmatrix}, \qquad b_2 = -1.$$

For $\boldsymbol{x} = (2, 0)$, multiplying by the first weight matrix and adding its bias gives

$$\begin{aligned} \boldsymbol{a}_1(\boldsymbol{x}) &= \begin{pmatrix} 2 & 0 \end{pmatrix} \begin{pmatrix} 1 & 2 & -1 \\ -1 & 0 & 1 \end{pmatrix} + \begin{pmatrix} 0 & 1 & 0 \end{pmatrix} = \begin{pmatrix} 2 & 5 & -2 \end{pmatrix}, \\ \boldsymbol{\phi}_1(\boldsymbol{x}) &= \mathbf{ReLU}(\boldsymbol{a}_1(\boldsymbol{x})) = \begin{pmatrix} 2 & 5 & 0 \end{pmatrix}. \end{aligned}$$

The final layer therefore produces

$$f_{\boldsymbol{\theta}}(\boldsymbol{x}) = \boldsymbol{\phi}_1(\boldsymbol{x})\boldsymbol{W}_2 + b_2 = 2 \cdot 2 - 5 + \tfrac{1}{2} \cdot 0 - 1 = -2. \tag{51}$$

(b) Regression. For regression, the output itself is the prediction. With target $y = 1$, the squared loss is therefore

$$\ell_{\mathrm{sq}}(f_{\boldsymbol{\theta}}(\boldsymbol{x}), 1) = (-2 - 1)^2 = 9.$$

(c) Classification. For binary classification, we first turn the score into a probability:

$$p_{\boldsymbol{\theta}}(\boldsymbol{x}) = \sigma(f_{\boldsymbol{\theta}}(\boldsymbol{x})) = \sigma(-2) = \frac{1}{1 + e^2} \approx 0.1192.$$

This is below $1/2$, so the predicted label is zero. Since the observed label is one, the logistic loss is

$$\ell_{\log}(p_{\boldsymbol{\theta}}(\boldsymbol{x}), 1) = -\log p_{\boldsymbol{\theta}}(\boldsymbol{x}) = \log(1 + e^2) \approx 2.1269.$$

4. Fixed features and learned features.

(a) Two inputs with the same hidden features. Multiplying a general input row by the first weight matrix and adding the bias gives

$$\boldsymbol{a}_1(\boldsymbol{x}) = (x_1 - x_2,\ 2x_1 + 1,\ -x_1 + x_2).$$

The hidden feature row is therefore

$$\boldsymbol{\phi}_1(\boldsymbol{x}) = \big(\mathrm{ReLU}(x_1 - x_2),\ \mathrm{ReLU}(2x_1 + 1),\ \mathrm{ReLU}(-x_1 + x_2)\big).$$

At the two inputs, we obtain

$$\begin{aligned} \boldsymbol{a}_1(\boldsymbol{x}_A) &= (0, -1, 0), \quad \boldsymbol{\phi}_1(\boldsymbol{x}_A) = (0, 0, 0), \\ \boldsymbol{a}_1(\boldsymbol{x}_B) &= (0, -3, 0), \quad \boldsymbol{\phi}_1(\boldsymbol{x}_B) = (0, 0, 0). \end{aligned}$$

The two inputs are different, but this feature transformation maps them to the same hidden row.

(b) Why changing the final layer cannot help. For any choice of $\boldsymbol{W}_2 \in \mathbb{R}^{3\times 1}$ and $b_2 \in \mathbb{R}$, keeping the first layer fixed gives

$$\begin{aligned} f_{\boldsymbol{\theta}}(\boldsymbol{x}_A) &= \boldsymbol{\phi}_1(\boldsymbol{x}_A)\boldsymbol{W}_2 + b_2 = b_2, \\ f_{\boldsymbol{\theta}}(\boldsymbol{x}_B) &= \boldsymbol{\phi}_1(\boldsymbol{x}_B)\boldsymbol{W}_2 + b_2 = b_2. \end{aligned}$$

The same bias would have to equal both 1 and 0, which is impossible. No choice of final-layer parameters can distinguish these inputs while their hidden feature rows remain identical.

(c) Changing a feature makes the targets attainable. After replacing the second hidden bias by 3, the new feature map is

$$\widetilde{\boldsymbol{\phi}}_1(\boldsymbol{x}) = \big(\mathrm{ReLU}(x_1 - x_2),\ \mathrm{ReLU}(2x_1 + 3),\ \mathrm{ReLU}(-x_1 + x_2)\big).$$

Its values at the two inputs are now different:

$$\widetilde{\boldsymbol{\phi}}_1(\boldsymbol{x}_A) = (0, 1, 0), \qquad \widetilde{\boldsymbol{\phi}}_1(\boldsymbol{x}_B) = (0, 0, 0).$$

Choose

$$\boldsymbol{W}_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \qquad b_2 = 0.$$

The final layer selects the second hidden feature, giving

$$\widetilde{\boldsymbol{\phi}}_1(\boldsymbol{x}_A)\boldsymbol{W}_2 + b_2 = 1, \qquad \widetilde{\boldsymbol{\phi}}_1(\boldsymbol{x}_B)\boldsymbol{W}_2 + b_2 = 0.$$

Thus changing the first layer makes it possible to fit both targets exactly. Adjusting only the final layer could not achieve this with the original feature map.

5. ReLU at zero. Since $\mathrm{ReLU}(0) = 0$, the difference quotient is

$$\frac{\mathrm{ReLU}(0 + h) - \mathrm{ReLU}(0)}{h} = \begin{cases} 0, & h < 0, \\ 1, & h > 0. \end{cases} \tag{52}$$

Its limit from the left is zero, while its limit from the right is one. The two limits disagree, so ReLU is not differentiable at zero. The computational convention $\mathrm{step}(0) = 0$ does not alter these unequal limits.

6. Gradient descent from two initialisations.

(a) Derivatives and update directions. The squared-loss formula (44) in the lecture notes is

$$\mathcal{L}_{\text{sq}}(\theta) = \begin{cases} 2(\theta+3)^2, & \theta \le -1, \\ 2(\theta+1)^2 + 8, & -1 < \theta < 1, \\ 2(\theta-3)^2 + 8, & \theta \ge 1. \end{cases}$$

Differentiating each quadratic on its open interval gives

$$\mathcal{L}'_{\text{sq}}(\theta) = \begin{cases} 4(\theta+3), & \theta < -1, \\ 4(\theta+1), & -1 < \theta < 1, \\ 4(\theta-3), & \theta > 1. \end{cases}$$

A gradient-descent update subtracts $\eta\mathcal{L}'_{\text{sq}}(\theta)$, where $\eta > 0$. Consequently,

θ	$\mathcal{L}'_{\text{sq}}(\theta)$	Effect of the update
-4	-4	increases θ
0	4	decreases θ
4	4	decreases θ

(b) Two gradient-descent runs. For the squared loss, the update is

$$\theta^{(t+1)} = \theta^{(t)} - 0.2\mathcal{L}'_{\text{sq}}\big(\theta^{(t)}\big).$$

Starting from $\theta^{(0)} = -2$, we use the derivative $4(\theta+3)$. The first two updates give

$$\theta^{(1)} = -2 - 0.2 \cdot 4 = -2.8,$$
$$\theta^{(2)} = -2.8 - 0.2 \cdot 0.8 = -2.96.$$

All three values are less than -1, so the same expression for the loss applies throughout. Their losses are

$$\mathcal{L}_{\text{sq}}(-2) = 2, \qquad \mathcal{L}_{\text{sq}}(-2.8) = 0.08, \qquad \mathcal{L}_{\text{sq}}(-2.96) = 0.0032.$$

Starting from $\theta^{(0)} = 2$, instead, we use the derivative $4(\theta-3)$. We obtain

$$\theta^{(1)} = 2 - 0.2 \cdot (-4) = 2.8,$$
$$\theta^{(2)} = 2.8 - 0.2 \cdot (-0.8) = 2.96.$$

These values remain greater than 1, and their losses are

$$\mathcal{L}_{\text{sq}}(2) = 10, \qquad \mathcal{L}_{\text{sq}}(2.8) = 8.08, \qquad \mathcal{L}_{\text{sq}}(2.96) = 8.0032.$$

For the first run, the update formula gives

$$\theta^{(t+1)} + 3 = \theta^{(t)} + 3 - 0.8\big(\theta^{(t)} + 3\big) = 0.2\big(\theta^{(t)} + 3\big).$$

Thus every iterate remains between -3 and the starting value -2, where this formula applies. The distance from -3 shrinks by a factor of 0.2 at each update, so the iterates approach -3 and the loss approaches zero.

For the second run, similarly,

$$\theta^{(t+1)} - 3 = 0.2\big(\theta^{(t)} - 3\big).$$

Every iterate remains between the starting value 2 and 3. The iterates approach 3, where the loss is eight. In both runs the squared distance from the respective minimum decreases at every update, so the loss decreases too. The second run nevertheless approaches a local minimum whose loss exceeds the global minimum value of zero. It is therefore a counterexample to the claim.

7. Constructing a function with ReLU.

(a) The piecewise formula. The breakpoints are $-1, 0, 1$. For $x \le -1$, all three ReLUs are zero. For $-1 < x \le 0$, only the first contributes a nonzero term, giving $2(x+1)$. For $0 < x < 1$, the first two give $2(x+1) - 4x = 2(1-x)$. Finally, when $x \ge 1$, their sum is $2(x+1) - 4x + 2(x-1) = 0$. Thus

$$U(x) = \begin{cases} 0, & x \le -1, \\ 2(x+1), & -1 < x \le 0, \\ 2(1-x), & 0 < x < 1, \\ 0, & x \ge 1. \end{cases}$$

The adjacent formulas agree at the breakpoints. In particular, $U(0) = 2$.

(b) Network parameters. The hidden layer must produce $\mathrm{ReLU}(x+1)$, $\mathrm{ReLU}(x)$ and $\mathrm{ReLU}(x-1)$. The final layer then combines them with weights $2, -4, 2$. Consequently, one suitable parameter choice is

$$\boldsymbol{W}_1 = \begin{pmatrix} 1 & 1 & 1 \end{pmatrix}, \quad \boldsymbol{b}_1 = \begin{pmatrix} 1 & 0 & -1 \end{pmatrix}, \quad \boldsymbol{W}_2 = \begin{pmatrix} 2 \\ -4 \\ 2 \end{pmatrix}, \quad b_2 = 0.$$

8. Correct labels, probabilities and loss.

(a) Scaling the XOR scores and probabilities. Keep the first-layer parameters fixed and multiply both final-layer weights and the final bias by α. This gives

$$\boldsymbol{W}_2 = \begin{pmatrix} \alpha \\ \alpha \end{pmatrix}, \qquad b_2 = -\frac{\alpha}{2}. \tag{53}$$

The resulting score is

$$f_{\boldsymbol{\theta}(\alpha)}(\boldsymbol{x}) = \alpha\big(\mathrm{ReLU}(x_1 - x_2) + \mathrm{ReLU}(x_2 - x_1) - \tfrac{1}{2}\big). \tag{54}$$

At the inputs $(0,0), (1,0), (0,1), (1,1)$, the scores are $-\alpha/2, \alpha/2, \alpha/2, -\alpha/2$, respectively. Since $\alpha > 0$, their signs are unchanged. The classifier therefore still predicts $(0,1,1,0)$, giving all four labels correctly.

At either class-zero input, the score is $-\alpha/2$, so the probability assigned to the observed label is $1 - \sigma(-\alpha/2) = \sigma(\alpha/2)$. At either class-one input, the score is $\alpha/2$, and the probability assigned to the observed label is again $\sigma(\alpha/2)$. Thus all four observations receive the same correct-label probability,

$$\sigma(\alpha/2) = \frac{1}{1 + e^{-\alpha/2}}.$$

This probability increases as α increases, while all four predicted labels remain unchanged.

(b) Logistic loss. Using the probabilities from part (a), each observation has the same loss:

$$\begin{aligned} \ell_{\log}(\sigma(-\alpha/2), 0) &= \log(1 + e^{-\alpha/2}), \\ \ell_{\log}(\sigma(\alpha/2), 1) &= \log(1 + e^{-\alpha/2}). \end{aligned} \tag{55}$$

There are four observations, so their summed loss is

$$\mathcal{L}_{\log}(\boldsymbol{\theta}(\alpha)) = 4 \log(1 + e^{-\alpha/2}) > 0. \tag{56}$$

For every finite α, the exponential is positive, so the argument of the logarithm is greater than one. The loss is therefore positive. As α increases without bound, that exponential approaches zero, and the loss approaches zero as well. The predicted labels remain correct throughout, while the probabilities assigned to them become increasingly confident.